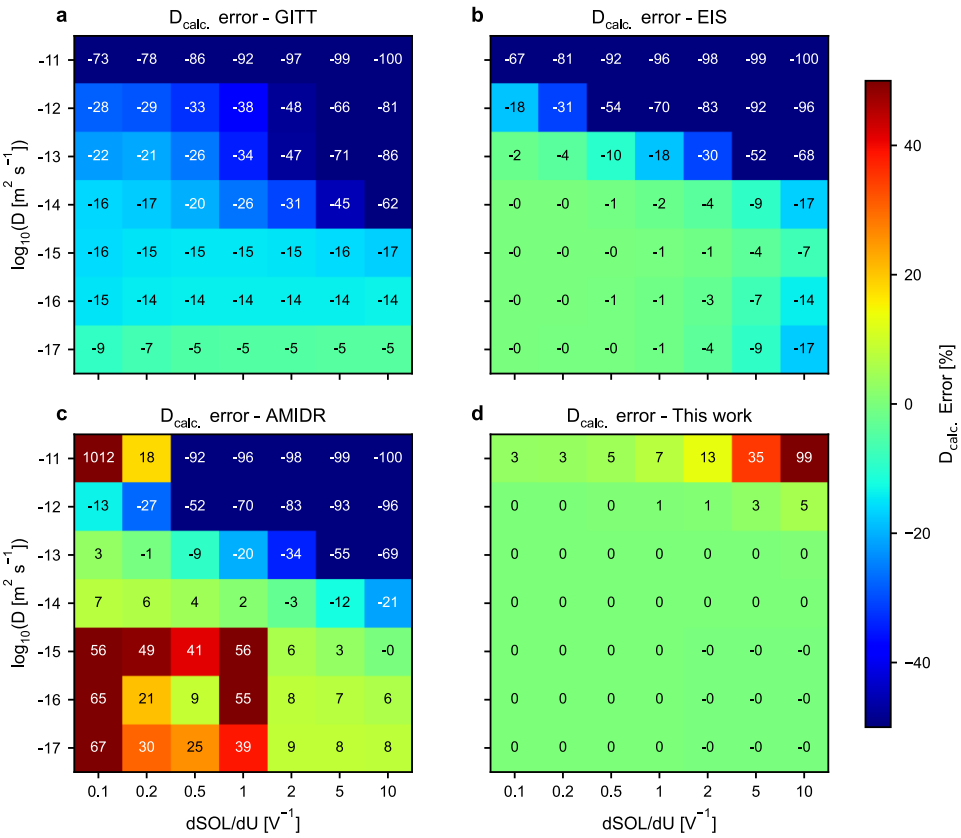
# Method for Measuring Solid-State Diffusivity: Part 1. Theoretical Introduction

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## Abstract

The reliable determination of solid-state lithium diffusivity is essential for parameterizing physics-based lithium-ion battery models. However, conventional methods misattribute electrolyte concentration polarization to solid-state diffusion, introducing systematic errors in the inferred diffusivity. This study introduces a resistance-decomposition method that explicitly accounts for electronic conduction, electrolyte transport, charge transfer, and solid-state diffusion within the Doyle–Fuller–Newman (DFN) framework. The diffusion coefficient is calculated from the residual polarization resistance after independently determining the other contributions and accounting for the equilibrium potential dependence on the state of lithiation. The method is evaluated using synthetic voltage data generated with the DFN model implemented in PyBaMM. The results demonstrate accurate recovery of the prescribed diffusivity across most of the investigated parameter space under idealized conditions.. These findings establish the theoretical basis and numerical feasibility of the proposed method while identifying key limitations that require experimental validation.




### Keywords

Diffusion, Li-ion, GITT, AMIDR, EIS, DFN, PyBaMM, NMC

# 1. Introduction

The solid-state lithium diffusion coefficient is a key parameter in physics-based lithium-ion battery models[1,2], governing concentration gradients within active-material particles and influencing electrode polarization, rate capability, and accessible state of lithiation. Reliable diffusivity determination is therefore essential for model parameterization and for interpreting the electrochemical behavior of battery materials.

Several electrochemical techniques are commonly used to determine solid-state diffusivity, including the galvanostatic intermittent titration technique (GITT)[3], electrochemical impedance spectroscopy (EIS)[4], intermittent current interruption (ICI)[5], and the Atlung Method for Intercalant Diffusion and Resistance (AMIDR)[6,7]. These methods infer diffusivity from voltage transients, impedance responses, or relaxation behavior using analytical or equivalent-circuit models. Their accuracy, however, depends on the validity of the underlying assumptions and the separation of solid-state diffusion from other polarization processes.

A central challenge is electrolyte transport in porous electrodes. Ionic conduction and the development of electrolyte concentration gradients both contribute to the measured voltage response. Because electrolyte concentration polarization can exhibit behavior similar to solid-state diffusion, neglecting it may lead to systematic errors in the inferred diffusion coefficient[8–10]. This limitation is particularly important when characterizing materials with relatively high solid-state diffusivity, for which electrolyte transport may constitute a substantial fraction of the measured polarization. The interpretation of electrochemical measurements must therefore account for electrolyte transport alongside interfacial kinetics and electronic resistance.

This work proposes a resistance-decomposition method for determining solid-state diffusivity within the Doyle–Fuller–Newman (DFN)[1] modeling framework. The method expresses the total effective polarization resistance as the sum of contributions from electronic conduction, electrolyte transport, charge transfer, and solid-state diffusion. The diffusion resistance is isolated by subtracting the independently determined contributions from the total resistance, after which the diffusion coefficient is calculated using an analytical expression for spherical particles and the thermodynamic relationship between capacity and equilibrium potential.

The method is evaluated using synthetic voltage data generated with the DFN model implemented in PyBaMM[11]. The numerical study examines diffusivity recovery during galvanostatic charge and discharge and across a systematic parameter space of diffusion coefficients and thermodynamic slopes. Particular attention is given to transient diffusion polarization and the sensitivity of the inversion when the residual diffusion resistance is small.

This first part establishes the theoretical formulation and numerical proof of concept. Experimental determination of the individual resistance contributions, uncertainty propagation, and validation using measured cell data are beyond its scope and will be addressed in subsequent work.

# 2. Theoretical Framework

## 2.1. Resistance decomposition and methodological principle

The proposed method determines the solid-state diffusion coefficient of an active material by decomposing the total polarization of an electrochemical cell into contributions associated with electrical conduction, electrolyte transport, charge transfer, and solid-state diffusion. The approach is formulated within the framework of the Doyle–Fuller–Newman (DFN) model, which provides a consistent basis for subsequent parameterization of physics-based lithium-ion cell models.

The motivation for this approach is that conventional solid-state diffusion determination methods often neglect electrolyte concentration polarization. Previous numerical investigations have demonstrated that this simplification can introduce substantial errors in the inferred solid-state diffusion coefficient, even for cells with low electrode loading. Moreover, conventional methods are intermittent and even with negligible electrolyte influence can introduce significant error, while the method described here provides continuous measurement during a single galvanostatic sweep once steady-state polarization is achieved. These observations motivate a method in which electrolyte transport and other independently characterized contributions are explicitly accounted for before solid-state diffusion is evaluated.

The method is based on assumptions:

1. The equilibrium potential of the active material, ($U$), is known as a function of its state of lithiation ($SOL$).
2. The active material consists of uniformly sized spherical particles with radius ($r$).
3. Lithium transport in the active material and electrolyte follows Fickian diffusion.
4. All relevant cell resistance contributions other than solid-state diffusion resistance can be independently determined.
5. Uniform reaction distribution.
6. The remaining polarization can be attributed to solid-state diffusion and used to calculate its diffusion coefficient.

Under these assumptions, the total cell resistance is represented as

$$R_{total} = R_{electric} + R_{electrolyte} + R_{CT} + R_D \quad (1)$$

where $R_{electric}$ represents electronic conduction and contact losses, $R_{electrolyte}$ accounts for electrolyte transport, $R_{CT}$ represents charge-transfer-related polarization, and $R_D$ represents the contribution associated with solid-state diffusion.

Equation (1) is an effective resistance representation of the voltage polarization under the specified operating conditions. In particular, the diffusion-related terms should be understood as time-dependent contributions that approach their steady-state values as the corresponding concentration gradients develop. Consequently, the individual contributions need not be fully developed during short current pulses.

## 2.2. Electrical resistance

The electrical resistance is decomposed into contributions from metallic cell components, contact interfaces, and electronic conduction through the electrode coating:

$$R_{electric} = R_{metal} + R_{contact} + R_{electrode} \tag{2}$$

The metallic contribution, $R_{metal}$, includes the current collectors and auxiliary conductive components connecting the electrodes to the cell terminals. Its determination generally requires a cell-specific electrical model or numerical calculation. This contribution is not included in the basic DFN implementation used in the present work and is therefore set to zero in the numerical validation.

The contact resistance between the electrode coating and current collector is expressed as

$$R_{contact} = \frac{\rho_{contact}}{A} \tag{3}$$

where $\rho_{contact}$ is the area-specific interfacial resistivity [$\Omega m^2$], and ($A$) is the electrode surface area [$m^2$].

The effective electronic resistance of the electrode coating is approximated by

$$R_{electrode} = \frac{l_{electrode}}{3A\sigma_{electrode}} \tag{4}$$

where $l_{electrode}$ is the electrode thickness [m] and $\sigma_{electrode}$ is the effective (apparent) electronic conductivity of the coating [$S.m^{-1}$]. The factor of three accounts for the assumed current distribution within the porous electrode, following the analogous treatment of ionic transport in electrode coatings described by Landesfeind et al[12].

Neither contact resistance nor effective electrode conductivity should necessarily be regarded as invariant material parameters. Both can depend on lithiation state[13] and ageing[14]. Contact resistance can be a substantial component of the measured electrical losses in practical cells.

For the DFN configuration employed here, $R_{metal}$ and $R_{contact}$ are not explicitly represented. The resulting electrical resistance is therefore determined by the electrode-conduction contribution and is equal to $3 \cdot 10^{-4}$ Ω in the simulated cell.

## 2.3. Electrolyte resistance, including concentration polarization

Electrolyte polarization comprises both the instantaneous response associated with ionic conduction and the time-dependent contribution associated with the development of concentration gradients. Neglecting the latter can lead to an incorrect assignment of electrolyte polarization to solid-state diffusion[8].

For a porous electrode and separator, the effective electrolyte resistance is approximated[15] as

$$R_{electrolyte} = \frac{\frac{M_{electrode} l_{electrode}}{3} + M_{separator} l_{separator}}{A\sigma_{total_electrolyte}} \tag{5}$$

where $l_{separator}$ is the separator thickness [m], $M$ is the McMullin number of each component, calculated as tortuosity divided by porosity[12] [-]. The factor of three is applied to the electrode contribution to represent the assumed current distribution through the porous coating[16]. No equivalent factor is applied to the separator because the ionic flux is constant through its thickness. The same area is used as DFN assumes no overhangs. The effective total electrolyte conductivity is defined through the reciprocal sum

$$\frac{1}{\sigma_{total_electrolyte}} = \frac{1}{\sigma_{electrolyte}} + \frac{1}{\sigma_{Warburg_electrolyte}} \tag{6}$$

where $\sigma_{electrolyte}$ represents the ionic conductivity [S.m$^{-1}$] and $\sigma_{Warburg_electrolyte}$ is the effective Warburg conductivity associated with electrolyte concentration polarization [S.m$^{-1}$]. Following the concentrated-solution formulation for Warburg conductivity[15], the latter is expressed as

$$\sigma_{Warburg_electrolyte} = \frac{D_{electrolyte} F^2 c}{2RT(TDF)(1-\tau_+)^2} \tag{7}$$

where $D_{electrolyte}$ is the electrolyte diffusion coefficient [m$^2$s$^{-1}$], $c$ is the electrolyte salt concentration [mol·m$^{-3}$], $F$ is the Faraday constant [C·mol$^{-1}$], $R$ is the universal gas constant [J·mol$^{-1}$·K$^{-1}$], $T$ is absolute temperature [K], $TDF$ is the thermodynamic factor[17] [-], and $\tau_+$ is the lithium-ion transference number [-].

The reciprocal combination in Equation (6) provides an effective representation of the combined ionic-conduction and concentration-polarization contributions. The first contribution is associated primarily with the immediate voltage response to current, whereas the second develops as electrolyte concentration gradients form under sustained current.

The electrolyte transport parameters depend on concentration and temperature. Consequently, their values can vary spatially during operation. For sufficiently small concentration gradients, they may be approximated as locally constant, which supports the simplified resistance representation adopted here.

Equation (5) also assumes an approximately uniform current distribution within the porous electrode. This assumption is most appropriate for thin, low-loading electrodes under kinetically limited conditions[16].

## 2.4. Charge-transfer resistance

The charge-transfer contribution is represented using the conventional interpretation of interfacial electrochemical kinetics. In practice, the experimentally identified contribution may incorporate effects beyond an idealized charge-transfer resistance. For the present method, the essential requirement is that the contribution be independently characterized with sufficient accuracy.

For spherical particles, the total charge-transfer resistance is expressed as

$$R_{CT} = \frac{RT}{Fi_0} \cdot \frac{r}{3V} \tag{8}$$

where $i_0$ is the exchange-current density [A·m$^{-2}$], $r$ is the particle radius [m], and $V$ is the total active-material volume [m$^3$]. The factor of three follows from the surface-area-to-volume ratio of

a sphere. The formulation assumes that the exchange-current density and the relevant particle geometry are defined consistently for the material and operating state being considered. It is correct only for linear Butler-Volmer regime.

As with other resistance contributions, charge-transfer resistance may depend on lithiation state, temperature, and material properties[4,18]. Such dependencies should be retained when applying the method.

## 2.5. Solid-state diffusion resistance

The solid-state diffusion resistance is defined as the effective resistance associated with the overpotential generated by a concentration gradient within the active material under sustained galvanostatic operation. The formulation follows the analytical treatment developed by Ball et al. in their work on the Atlung Method for Intercalant Diffusion and Resistance (AMIDR)[7]. A related representation can be obtained from the spherical diffusion impedance discussed by Charbonneau et al[4].

For particles of a specified geometry, the steady-state diffusion resistance is written as

$$R_D = \frac{r^2}{B\frac{dQ}{dU}D} \tag{9}$$

where $D$ is the solid-state diffusion coefficient [$m^2s^{-1}$], $Q$ is the theoretical capacity of the active material in cell [C], $U$ is its potential [V], and $B$ is a geometry-dependent constant [-]. The derivative (dQ/dU) is defined using a capacity convention for which it is positive.

For spherical particles, $B$ = 15. The corresponding constants for planar and cylindrical geometries are 3 and 8, respectively. The present work considers only spherical particles. The pseudo-steady-state regime is reached after surpassing Fourier number, equal 0.2 for spheres, of time constant ($t$) defined as:

$$t = \frac{r^2}{D} \tag{10}$$

Rearranging Equation (9) gives the diffusion coefficient:

$$D = \frac{r^2}{B\frac{dQ}{dU}R_D} \tag{11}$$

The method therefore requires the solid-state diffusion resistance to be isolated from the total resistance. Combining Equations (1) and (11) gives

$$D = \frac{r^2}{B\frac{dQ}{dU}(R_{total} - R_{electric} - R_{electrolyte} - R_{CT})} \tag{12}$$

Equation (12) is the central inversion equation of the proposed method. It shows that diffusivity is determined from the residual resistance after the other contributions have been accounted for.

The diffusion resistance in Equation (9) represents the steady-state limit of the polarization associated with solid-state concentration gradients. During shorter operations, the concentration

profile may not have reached its steady-state form, and the measured diffusion contribution can therefore be smaller than the value predicted by Equation (9). This distinction is essential when selecting the duration of galvanostatic measurements.

The quantities *D*, *dQ/dU*, and, in some materials, *r* depend on *SOL*. This dependence can be especially pronounced in materials undergoing substantial volume changes, including silicon-based materials and graphite. The basic DFN implementation used here does not account for changes in particle radius during lithiation. The present validation adopts the same simplification; a state-dependent particle radius could be introduced in a subsequent extension.

## 2.6. Determination of total resistance from galvanostatic measurements

The total effective resistance is obtained from the difference between the measured electrode potential and the equilibrium potential at the corresponding *SOL*:

$$R_{total_charge} = \frac{U_{charge} - OCV_{charge}}{I_{charge}} \ (13a), \qquad R_{total_discharge} = \frac{U_{discharge} - OCV_{discharge}}{I_{discharge}} (13b)$$

Here, the charge current is positive and the discharge current is negative. The potentials $U_{charge}$ and $U_{discharge}$ denote the measured operating potentials, whereas $OCV_{charge}$ and $OCV_{discharge}$ denote the corresponding equilibrium potentials. In this notation, *OCV* refers to the equilibrium potential of the half-cell electrode under investigation, rather than necessarily the open-circuit voltage of a complete two-electrode cell. Charge and discharge *OCV* are distinguished as usually *OCV* measured during charge cycle is higher than one measured during discharge. Such differences primarily stem from incomplete relaxation (a source of error) and potential hysteresis, which can be addressed by differentiating between the two OCVs. Because the model employed here does not account for hysteresis and utilizes fully relaxed OCV data, both OCV values and, consequently, the resulting total resistances ($R_{\text{total}}$) are treated as equal. Nonetheless, this distinction is maintained to accurately reflect real-world cells, which typically exhibit potential hysteresis and imperfect relaxation.

For the resistance interpretation to be valid, the equilibrium potential must correspond to the same *SOL* and temperature as the operating measurement, and the current convention must be applied consistently. The resulting quantity is an effective polarization resistance at the specified operating condition.

The charge and discharge estimates need not be identical. Differences can arise from state dependence, kinetic asymmetry, incomplete relaxation, and deviations from the idealized resistance representation. Agreement between the two directions is therefore a useful consistency check rather than a requirement imposed by the measurement equations.

Once the $R_{total}$ has been obtained, Equation (12) gives the corresponding diffusion coefficient.

## 2.7. Cell configuration and methodological limitations

The method is formulated for a single active material. If two active materials contribute simultaneously to the measured response, the individual diffusion resistances cannot generally be separated from the total resistance without additional information. Accordingly, the proposed

approach requires a half-cell configuration in which the electrode of interest can be characterized independently.

Consistent with the AMIDR methodology, a reference electrode is desirable to minimize the influence of the counter electrode on the measured potential. Low electrode loading is also beneficial because it reduces electrolyte polarization and other resistance contributions relative to the signal associated with solid-state diffusion. Nevertheless, explicitly accounting for electrolyte resistance may allow the approach to be applied to higher-loading electrodes, as electrodes representative of commercial cells.

The method has an important error-compensation characteristic. If a resistance contribution is omitted or incorrectly estimated, the residual $R_D$ absorbs the resulting discrepancy. This can preserve the agreement between a parameterized model and a measured voltage response, but the inferred diffusion coefficient will no longer represent the intrinsic material diffusivity. The method is therefore only as reliable as the independently determined resistance contributions and equilibrium-potential relationship.

The assumption of Fickian diffusion also requires qualification. Some commonly used electrode materials, particularly those undergoing phase transitions, may not follow a simple Fickian description over their full lithiation range. The present method nevertheless uses the Fickian diffusion framework implemented in the DFN model. A state-dependent effective diffusivity may reproduce some of the corresponding voltage response, but this possibility remains a hypothesis requiring further validation, particularly at higher currents.

The present paper therefore establishes the theoretical basis and numerical proof of concept for the method. Detailed experimental procedures for independently determining the resistance contributions and validating the resulting diffusivities are outside the scope of this part and are reserved for subsequent work.

# 3. Numerical Validation Methodology

## 3.1. DFN model and simulation configuration

The method was validated using synthetic voltage data generated with the Doyle–Fuller–Newman (DFN) model implemented in PyBaMM. The DFN framework was selected because the proposed method is intended for parameterization of physics-based cell models and because it allows the true input parameters to be compared directly with the values recovered by the inversion procedure.

The parameter set used in the simulations was adopted from previous work[8–10], enabling direct comparison with the corresponding benchmark error maps. The principal parameters are summarized in Table 1.

**Table 1.** Parameters used in the models copied from previous work[8]. The parameters with citations are taken from the cited work. In the case of $cm^2$, these refer to the surface area of the electrode, not to the surface area of the active material. Electrode thickness is limited by NMC diameter, in real example might be higher due to particle size distribution.

| **Parameter** | **Value** |
|---|---|
| Temperature | 25°C |
| Electrode surface area | 1 $cm^2$ |
| Electrode loading | 1.0 mg NMC ·$cm^{-2}$ |
| Active material content | 93 % wt. |
| Coating conductivity | 100 S·$m^{-1}$ |
| Coating contact resistance | 0 Ω.$cm^2$ |
| Electrode thickness | 9 µm |
| Electrode porosity | 72.549 % |
| Electrode tortuosity[12] | 1.5 |
| Separator thickness | 50 µm (double Celgard 2500) |
| Separator porosity | 55 % |
| Separator tortuosity[12] | 2.5 |
| Electrolyte, model[17] | 1M $LiPF_6$ EC:EMC 3:7 wt. |
| NMC Size (diameter)[4] | 9 µm |
| NMC crystallographic density[19] | 4.7828 g·$cm^{-3}$ |
| NMC theoretical capacity | 278 mAh/g |
| Double layer capacitance ($C_{DL}$)[4] | 0.14 mF·$cm^{-2}$ |
| Charge transfer resistance ($R_{CT}$)[4] | 18.29 Ω·$cm^{-2}$ |

The model was configured as a half-cell, with the potential of the electrode under investigation referenced to the electrolyte potential between two thin separators. The electrode mass loading was set to 1 mg·$cm^{-2}$, corresponding to approximately 0.2 mAh·$cm^{-2}$. With such extremely low loading, coating thickness is defined by particle diameter, so porosity is high.

## 3.2. Validation strategy

The numerical validation comprised two complementary tests.

The first test evaluated the recovery of the diffusion coefficient from simulated constant-current charge and discharge trajectories for NMC111 using the previously established parameter set. The objective was to determine how the inferred diffusivity depends on current magnitude, current direction, and *SOL* for a material representing real properties.

The second test evaluated the method over a systematic parameter space by varying the solid-state diffusion coefficient and the thermodynamic slope (*dQ/dU*). The resulting relative-error map was used to identify parameter regimes in which the inversion accurately recovers the known diffusivity and regimes in which numerical or resistance-subtraction errors become significant.

Together, these tests examine both the behavior of the method during practical galvanostatic trajectories and its parameter-recovery performance under controlled idealized conditions.

## 3.3. Charge and discharge simulations

Constant-current simulations were performed over a range of charge and discharge rates. The lowest simulated current was 1/240 C, where the C-rate was defined relative to theoretical active-material capacity. Lower currents were not used because the resulting overpotential was too small for reliable numerical extraction of the resistance.

Discharge was simulated at rates up to 1 C. Charge currents were restricted to 1/100 C because simulations initiated at *SOL* = 1 could not be initiated reliably at higher currents, due to high resistance ($R_D$ and $R_{CT}$) at this state.

At each simulated state, the operating potential and equilibrium potential were used to calculate the total effective resistance according to Equations (12a) and (12b), but as basic PyBaMM does not include hysteresis and known *OCV* dependency on *SOL* was used, so charge and discharge *OCV* are equal. The independently specified electrical, electrolyte, and charge-transfer contributions were then subtracted to obtain the residual diffusion resistance. Finally, Equation (11) was used to calculate the inferred diffusion coefficient.

The inferred values were compared with the known diffusivity prescribed in the DFN model. This procedure was applied throughout each simulated trajectory, allowing the development of the diffusion-related polarization and the associated error to be evaluated as functions of *SOL*.

## 3.4. Parameter-space sweep

The second validation test used 49 simulated pulses generated from seven values of the NMC solid-state diffusion coefficient and seven values of the *dSOL/dU*. Both parameters were held constant during each individual pulse, allowing their effects on the inferred diffusivity to be isolated.

The diffusion coefficient, $D_{NMC}$, was varied from $10^{-11}$ to $10^{-17}$ $m^2s^{-1}$. This range spans relatively fast solid-state transport and extremely slow diffusion regimes reported or considered for lithium-ion electrode materials.

The thermodynamic slope was expressed using the normalized theoretical capacity:

$$\frac{dSOL}{dU} = \frac{dQ}{dU} \cdot \frac{1}{Q} \tag{14}$$

The parameter *dSOL/dU* was varied across the range expected for NMC electrodes based on previous work[8].

The initial *SOL*, measurement duration, current direction and current magnitude were kept constant across all tests at 50%, $10^6$ s, discharge and $2 \cdot 10^{-7}$ A (approximately C/1000), respectively. Although these parameters do not reflect practical operational conditions, they are tailored specifically to the requirements of the method. This is particularly crucial at high $R_D$, where an extended duration is necessary for the resistance to fully develop, while the current must remain low to prevent an excessive overpotential. Notably, while up to a 28% change in SOL occurs during the $10^6$ pulse, holding *D* and *dSOL/dU* constant prevents parameter drift, thereby enabling the numerical validation of the steady-state equation.

The particle radius was fixed at $4.5 \cdot 10^{-6}$ m. Because the characteristic diffusion timescale equals $r^2/D$, the present parameter sweep does not independently investigate different particle radii. The resulting validation therefore applies directly to the selected radius. Comparisons at different radii require appropriate rescaling and consistent treatment of the other model parameters.

## 3.5. Error calculation

The relative error in the inferred diffusion coefficient was calculated as

$$Error = \frac{D_{Calculated}}{D_{PyBaMM}} - 1 \tag{15}$$

where $D_{calculated}$ is obtained using the proposed method and $D_{PyBaMM}$ is the known diffusivity used to generate the synthetic data.

An error of zero indicates exact recovery of the input diffusion coefficient. Positive values indicate overestimation, whereas negative values indicate underestimation. While standard relative error is asymmetric, bounded at -100% for underestimation and unbounded for overestimation, it is retained here for direct comparison with prior benchmark studies[8–10].

## 3.6. Scope of the numerical benchmark

The simulations represent an idealized best-case validation. All parameters required to calculate the diffusion coefficient were taken from the same model configuration used to generate the voltage response. Consequently, the exercise isolates the mathematical consistency and numerical behavior of the proposed inversion under the selected assumptions.

Experimental measurements will introduce additional sources of uncertainty, including particle-size distributions, nonuniform coatings, component misalignment, incomplete electrolyte impregnation, gas bubbles (incomplete wetting), and possible disturbances associated with the reference and counter electrodes. In addition, experimental uncertainties in the independently measured resistance contributions and equilibrium potential will propagate into the calculated diffusivity.

These effects are not quantified in the present benchmark. The numerical results should therefore be interpreted as a theoretical estimate of performance under idealized conditions, rather than a prediction of experimental accuracy.

# 4. Results and Discussion

## 4.1. Recovery of diffusivity during galvanostatic charge and discharge

Figure 1 presents the simulated voltage responses and the corresponding diffusivities calculated using the proposed method during galvanostatic charge and discharge. The lowest current investigated was 1/240 C, whereas discharge rates extended to 1 C. The charge rate was limited to 1/100 C because of numerical difficulties when charging from *SOL* = 1 at higher currents due to high resistance.

The calculated diffusivities show that the method does not recover the correct value immediately after the beginning of a current step. This behavior is consistent with the theoretical formulation: Equation (9) describes the steady-state diffusion resistance, whereas the solid-state concentration gradient requires time (0.2*t*) to develop and also continuously changes with *SOL*. During the initial transient, the diffusion resistance is therefore not fully represented by its steady-state value, and the inversion can produce substantial errors.

This effect is visible in the high-rate discharge trajectory and in the two low-rate charge trajectories. During discharge at 1 C, the error remains significant over an appreciable portion of the trajectory. During charging at 1/100 and 1/240 C, the error can also remain elevated because the simulated diffusivity is extremely small, resulting in a relatively long characteristic diffusion timescale despite the low current.

The results emphasize that the duration of the galvanostatic measurement must be considered relative to the characteristic solid-state diffusion timescale. A resistance inferred before the concentration profile has sufficiently developed cannot be interpreted directly as the steady-state resistance defined in Equation (9).

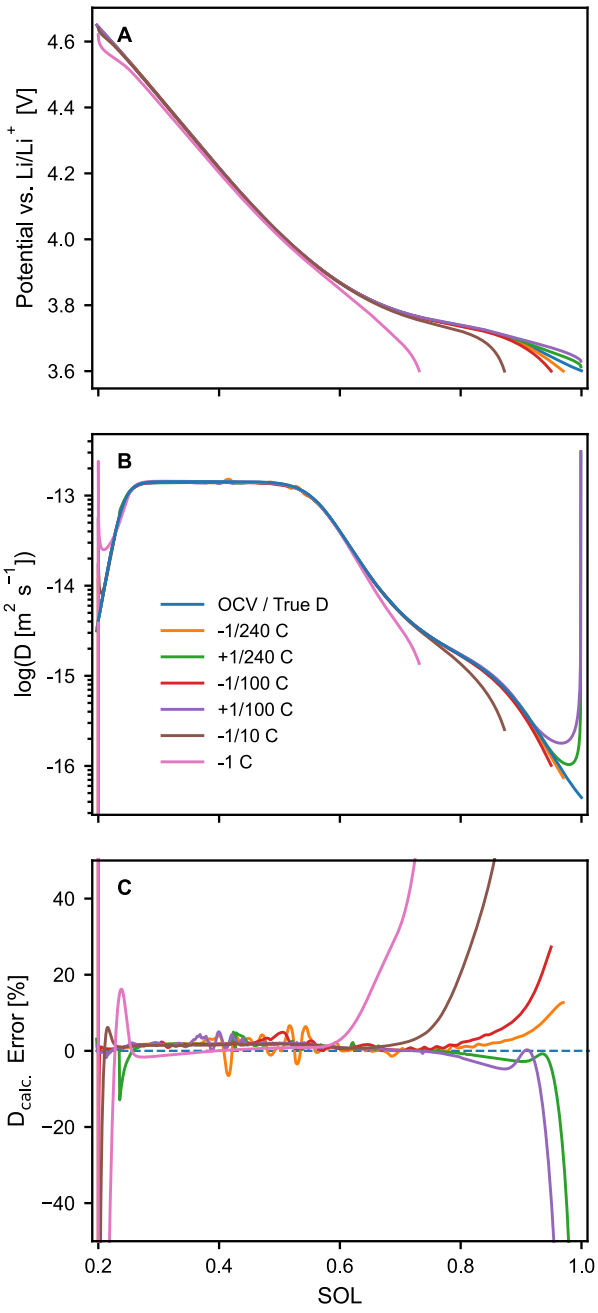



**Figure 1.** (A) OCV of NMC as a function of SOL, together with the corresponding potential profiles obtained at the investigated C-rates. (B) NMC chemical diffusion coefficient, together with the values obtained at the investigated C-rates with the method. (C) Error in $D_{calc.}$. The dashed horizontal line denotes zero deviation, while the ordinate is limited to ±50% to emphasize the practically relevant error range.

At the lowest currents, the general tendency is toward lower error once the diffusion polarization has developed. However, isolated error spikes occur at the smallest overpotentials. These spikes demonstrate a competing limitation: reducing the current reduces the magnitude of the polarization signal and can make the resistance subtraction numerically ill-conditioned. The useful operating window is therefore not determined by current magnitude alone. It also depends on the diffusivity, thermodynamic slope, measurement duration, and accuracy with which the voltage response can be resolved.

## 4.2. High-lithiation limitations and premature termination of discharge

A second limitation becomes apparent during high-rate discharge. In the simulated NMC111 case, the 1 C discharge terminates at approximately *SOL* = 0.75, rather than reaching full lithiation. The termination is associated with the sharp reduction in the prescribed diffusion coefficient at high *SOL*.

As the material approaches high lithiation, the falling diffusivity increases the resistance to further intercalation. Under high current, the concentration profile can become strongly nonuniform, and the surface region can approach its limiting composition while the interior remains less lithiated. The resulting polarization increases and can eventually cause the simulated electrode potential to reach its voltage cutoff.

This behavior is reinforced by the strong *SOL* dependence of diffusivity. As the surface region becomes increasingly difficult to lithiate, further intercalation produces additional polarization, restricting the accessible lithiation range at high current.

At lower currents, the smaller imposed flux reduces the concentration gradients and allows the simulation to access more extreme *SOL* values. However, very low currents also reduce the measurable overpotential, making the inversion more sensitive to numerical precision and measurement noise.

The results therefore identify a practical trade-off between sufficient polarization to resolve the resistance contributions and sufficiently low current to avoid premature termination of the trajectory. The intermediate current range between 1/100 and 1/10 C appears promising for further evaluation because it offers a compromise between polarization magnitude and access to a broad *SOL* window. This range should be regarded as a candidate operating window rather than a universally validated optimum.

The high-*SOL* behavior also illustrates the limitations of the present model assumptions. The DFN configuration used here does not account for particle-size evolution during lithiation, and the simulated behavior should not be interpreted as a complete description of degradation or structural changes in real NMC materials.

## 4.3. Error map across diffusivity and thermodynamic slope

Figure 2d presents the relative error in the calculated diffusion coefficient across the investigated parameter space. The map combines the results of the 49 synthetic pulses generated using seven diffusion coefficients and seven values of *dSOL/dU*.

For most investigated parameter combinations, the calculated diffusion coefficient closely matches the value used in the DFN simulation. The largest errors occur in the high-diffusivity regime, particularly for $D = 10^{-11}$ $m^2s^{-1}$ and $dSOL/dU > 1$ $V^{-1}$. In this regime, the calculated diffusivity exhibits errors reaching the two-digit percentage range, whereas most other parameter combinations show errors close to or equal zero. These deviations are attributed to simplifying assumptions underlying the analytical resistance equations (4–8) and the finite numerical precision of the PyBaMM solver.

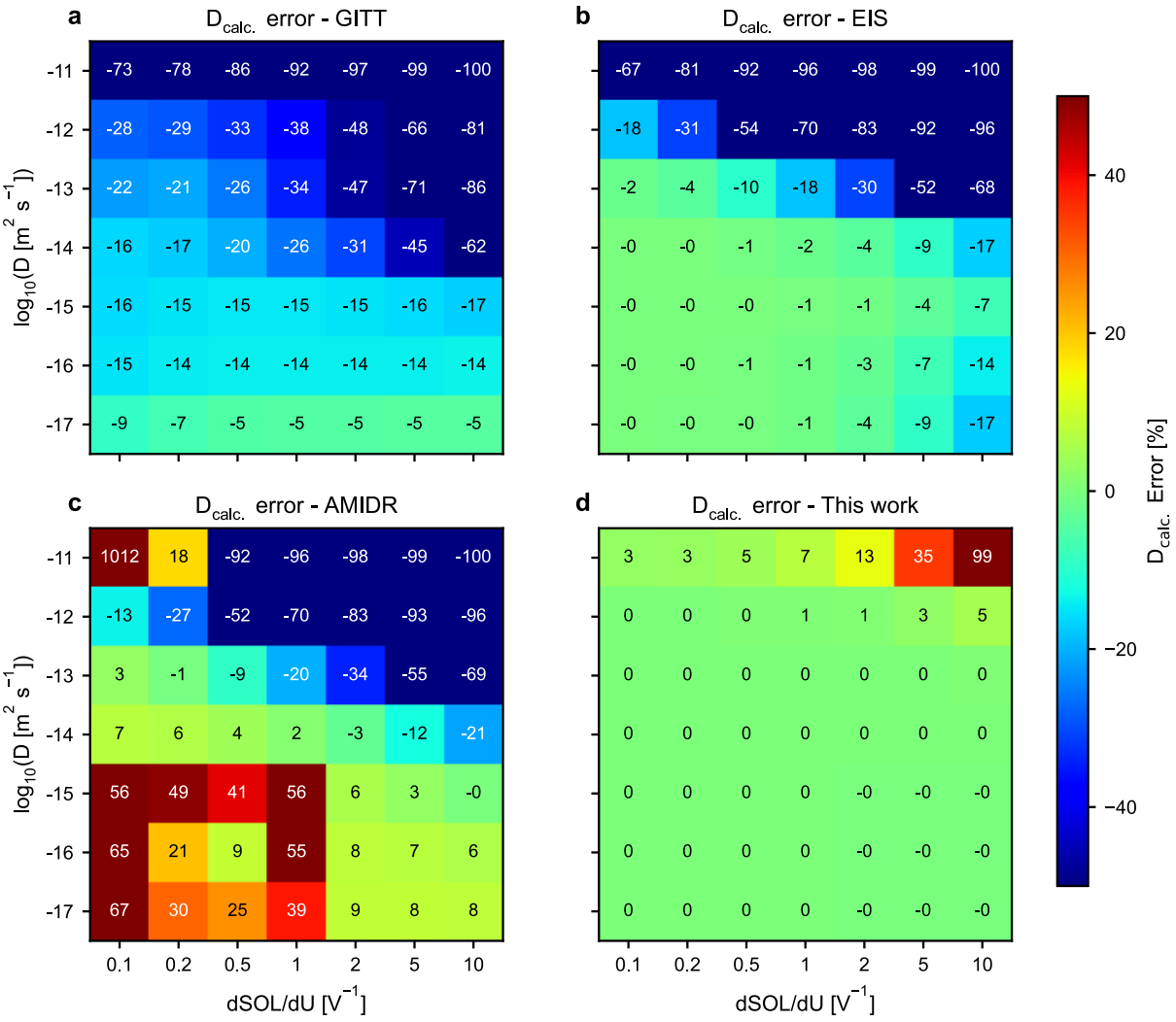



**Figure 2.** Comparison of error heatmaps for the *D* obtained via different determination methods. All methods use identical cell parameters across various *D* and *dSOL/dU* pairs in the PyBaMM DFN model. (a) GITT, adapted from Fig. 4 in Ref.[10]; (b) EIS, adapted from Fig. 10 in Ref.[8]; (c) AMIDR, adapted from Fig. 4 in Ref.[9]; and (d) the method proposed in this work.

This trend is consistent with the resistance decomposition. At high diffusivity, the steady-state solid-state diffusion resistance becomes small relative to the other resistance contributions. The method determines $R_D$ by subtracting those contributions from the total resistance. Even a small relative error in one of the larger terms can therefore become a substantial fraction of the residual resistance.

The sensitivity can be expressed directly by considering Equation (12). Method becomes increasingly sensitive when $R_D$ is small. An absolute resistance error that is negligible relative to

the total cell resistance can nevertheless be substantial relative to the diffusion resistance, producing a large relative error in the inferred diffusivity.

The error map should therefore be interpreted as a test of the inversion under the idealized conditions of the adopted simulation. It demonstrates that the resistance decomposition can recover the known diffusivity when the required inputs are internally consistent, while also identifying a fundamental sensitivity limit in regimes where the diffusion contribution is small.

## 4.4. Comparison with established diffusion determination methods

The parameter-space benchmark can be compared with previously reported synthetic validation results for established diffusion determination methods (Figure 2), including GITT, EIS-based approach and AMIDR. These comparisons are useful for identifying the regimes in which neglecting electrolyte transport produces substantial errors.

However, the comparison is not strictly like-for-like. The present method uses the known simulation parameters directly to calculate the resistance contributions, whereas the established methods use different assumptions and extraction procedures. The present results consequently represent an idealized parameter-recovery benchmark and should not be interpreted as a direct experimental-performance ranking.

Within this limitation, the proposed method demonstrates close recovery of the known diffusion coefficient across most of the investigated parameter space. Its principal advantage is the explicit accounting for electrolyte polarization rather than assigning the entire non-instantaneous voltage response to solid-state diffusion.

The remaining errors at high diffusivity are not evidence that electrolyte polarization has been neglected in the method. Instead, they reflect the sensitivity of residual-resistance extraction when the diffusion contribution is small relative to the other contributions. This distinction is important when interpreting the error map and comparing it with methods that neglect electrolyte diffusion altogether.

A rigorous quantitative comparison will require that all parameters used within the method are obtained experimentally.

## 4.5. Implications for practical parameterization

The numerical results support the use of the proposed method as a framework for determining effective solid-state diffusion coefficients for DFN parameterization. The approach is particularly attractive when the electrolyte transport parameters, equilibrium potential, electronic resistance, and charge-transfer contribution are already required for the intended cell model.

Although the number of independently characterized quantities may initially appear to complicate the measurement, these quantities are also necessary for a comprehensive DFN parameterization. The proposed approach therefore reorganizes the characterization task rather than introducing entirely new parameter requirements.

The results also indicate that the method cannot be expected to provide equally reliable diffusivities under all operating conditions. Measurements must be long enough for the diffusion-related polarization to develop, but not so weakly polarized that the residual resistance becomes

dominated by uncertainty. Similarly, operation at high currents can restrict the accessible SOL range and prevent determination of diffusivity in the most highly lithiated states. The practical accuracy of the method will ultimately depend on the precision of the independently determined resistance contributions.

## 5. Scope of the Present Study

This part establishes the theoretical formulation and numerical proof of concept of a resistance-decomposition method for solid-state diffusivity determination. The DFN simulations demonstrate that the method can recover the prescribed diffusion coefficient under idealized conditions and identify the regimes in which transient polarization, small residual diffusion resistance, and high-lithiation limitations compromise the inversion.

The study does not establish experimental accuracy or validate the independent measurement of every resistance contribution. Those questions require additional investigation, which will be covered in subsequent parts.

## 6. Conclusion

A resistance-decomposition method for determining solid-state lithium diffusivity has been formulated within the Doyle–Fuller–Newman framework. The method explicitly accounts for electronic conduction, electrolyte ionic transport and concentration polarization, and charge-transfer resistance, allowing the solid-state diffusion resistance to be isolated from the total measured polarization. The diffusion coefficient is subsequently calculated using an analytical expression for spherical particles and the thermodynamic slope of the active material.

Numerical validation using synthetic PyBaMM data demonstrates that the method can recover the prescribed diffusion coefficient across most of the investigated parameter space under idealized conditions. The results also identify two principal limitations: insufficient development of the solid-state concentration gradient during transient measurements and increased sensitivity to resistance-subtraction errors when the diffusion resistance is small relative to the other contributions. High-current operation can additionally restrict the accessible lithiation range, whereas very low currents can produce polarization signals that are too small for reliable resistance extraction.

The method's accuracy therefore depends on measurement duration, operating current, thermodynamic properties, and the independent characterization of the resistance contributions. The present results establish numerical consistency rather than experimental accuracy, since the required inputs were taken from the model used to generate the synthetic data.

Overall, the proposed framework provides a basis for incorporating electrolyte polarization explicitly into solid-state diffusivity determination and for improving the consistency of electrochemical characterization with DFN model parameterization. Experimental validation and quantitative assessment of parameter uncertainty are required to establish its practical applicability.

## Data availability statements

All the code used in preparing this study is available in the GitHub repository: https://github.com/MarcinK00/Diffusion_determination

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